

# Linyi (Tiger) Hou

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## EDUCATION

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Ph.D. in Aerospace Engineering, University of Illinois Urbana-Champaign  
B.S. in AE, Minor in CompE, University of Illinois Urbana-Champaign

Jun 2021 - present  
Aug 2017 - May 2021

## SKILLS

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|                       |                                                            |
|-----------------------|------------------------------------------------------------|
| Programming Languages | Python, C/C++, MATLAB, Javascript                          |
| Libraries/Frameworks  | numpy   astropy   scipy   pybind11, Eigen   OpenMP, NodeJS |
| Tools/Databases       | Git, GitHub, Slurm, MongoDB, ADQL                          |
| Operating Systems     | Linux, MacOS, Windows                                      |

## HIGHLIGHTS FOR SWE POSITION

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### Software Development for Graduate Research

- ❖ Developed software simulations and tools for navigation and astrodynamics research using Python/C++.
- ❖ Familiar with using Git to manage codebases and collaborate with colleagues.
- ❖ Familiar with cluster computing and job scheduling using Slurm.

### SWE Internship

- ❖ Wrote Python scripts to automate weekly builds and testing in Subversion.
- ❖ Migrated an avionics data interpreter from 16-bit to 32-bit using C++ and Python.
- ❖ Added functionality to a Perl script that generates HTML pages for data visualization.

### Personal Web Development Project

- ❖ Developed a puzzle hunt website using Node + Express with MongoDB.

## WORK EXPERIENCE

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### University of Illinois Urbana-Champaign

*Teaching Assistant (AE 202, AE 443, AE 483)*

Jun 2021 - present

- ❖ Worked with 2-5 faculty and staff to teach lessons, hold office hours, and improve curriculum for courses in flight mechanics, systems engineering, and UAV controls.
- ❖ Assisted in developing code and instructions for a UAV lab based on C and Python.

### Collins Aerospace

*Software Engineering Intern*

May 2019 - Aug 2019

- ❖ Migrated an avionics data interpreter from 16-bit to 32-bit using C++ and Python.
- ❖ Automated builds in TortoiseSVN using Python, reducing need for manual integration.

## RESEARCH

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### **Astrodynamics and Planetary Exploration Group (APEX)**

Advisor: Dr. Siegfried Eggli

#### *Interstellar Navigation and Star Identification*

Sep 2024 - present

- ❖ Developing star identification techniques for interstellar travel at relativistic speeds by combining traditional algorithms with clustering based on stellar classification.

#### *Celestial Navigation to Supplement DSN Tracking*

Jul 2024 - present

- ❖ Collaborating with four colleagues to assess feasibility of using celestial navigation to meet cruise phase uncertainty requirements in heliocentric and cislunar space.
- ❖ Designing sensor tasking/scheduling algorithms with novel considerations for covariance propagation and visibility constraints over the time domain.
- ❖ Implemented spherical harmonics for an open source orbit propagator in C++.

#### *Lost in Space and Time Navigation using Variable Stars*

Nov 2022 - present

- ❖ Developed an autonomous optical navigation technique based on light curve analysis of variable stars which does not require prior knowledge of position or time.
- ❖ Simulated star tracker observations and performed Monte Carlo analysis to assess the feasibility of reestablishing Earth pointing via the proposed method.

### **Aerospace Mission Analysis Laboratory**

Advisor: Dr. Zachary R. Putnam

#### *Lost in Space X-ray Pulsar Navigation (XNAV)*

Jun 2021 - present

- ❖ Developed algorithms to resolve spacecraft position and time from pulsar timing data.
- ❖ Created a photon time of arrival simulation tool in Python and validated results against the pulsar timing package [PINT](#) to within 50  $\mu$ s.
- ❖ Demonstrated the computational feasibility of (XNAV) in a lost in space scenario.

#### *Velocity-based Initial Orbit Determination*

Aug 2019 - Apr 2024

- ❖ Developed a novel initial orbit determination (IOD) technique using sequential range-rate measurements to distant stars.
- ❖ Assessed the performance of velocity IOD methods using Monte Carlo simulation.

## HONORS AND AWARDS

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### **Yee Fellowship**

2021, 2024

Grainger College of Engineering, University of Illinois Urbana-Champaign

### **Aerospace Engineering Alumni Advisory Board Fellowship**

2024

Department of Aerospace Engineering, University of Illinois Urbana-Champaign

### **Best Student Paper, 46<sup>th</sup> AAS GN&C Conference**

2024

American Astronautical Society (AAS) Rocky Mountain Section

### **David and Catherine Thompson Space Technology Scholarship**

2020

American Institute of Aeronautics and Astronautics (AIAA)